

Module 9 – Application Debugging

These notes are written based on my understanding after studying the Application Debugging module. I have summarized the important concepts in simple language as I understood them during learning.

1. What I Learned in this Module

I learned how to debug Angular applications using Chrome DevTools and Visual Studio Code. I also learned how to inspect HTML elements, debug TypeScript code, use breakpoints, analyze variables, inspect Angular services and understand state management.

2. Why Debugging is Important

Debugging is the process of identifying, analyzing and fixing errors in an application. It helps developers improve code quality, solve unexpected issues and build reliable applications.

3. Debugging Angular using Chrome DevTools

To debug an Angular application, I first run the application using 'ng serve' and open it in Chrome. By pressing F12, Chrome DevTools opens. From there I can inspect HTML elements, CSS styles and JavaScript execution.

4. Inspecting Elements (DOM Tree)

The Elements panel displays the complete DOM tree of the webpage. It helps identify incorrect HTML structure, missing elements, styling issues and layout problems.

5. Sources Panel and Breakpoints

The Sources panel allows debugging JavaScript and TypeScript code. By placing breakpoints, execution pauses at specific lines, making it easier to inspect variables, follow the call stack and understand the program flow.

6. Debugging TypeScript with Source Maps

Angular uses TypeScript, but browsers execute JavaScript. Source maps allow developers to debug the original TypeScript code instead of the compiled JavaScript, making debugging much easier.

7. Debugging Angular in Visual Studio Code

I learned how to install Angular-related extensions, configure launch.json, attach the VS Code debugger and debug the application directly from the editor.

8. Breakpoints, Watches and Variables

Breakpoints stop program execution at selected lines. Watches help monitor variable values during execution. The Variables window displays current values and helps understand how data changes while debugging.

9. Debugging State Management

I learned that Angular applications often use services or NgRx Store to manage data. During debugging, developers can inspect service data and application state to identify incorrect values or unexpected behavior.

10. Good Debugging Practices

- Understand the problem before fixing it.
- Use breakpoints instead of too many console.log statements.
- Inspect variables carefully.

- Read error messages completely.
- Test the application after every fix.

11. Platforms and Related Skills

This module also highlighted the importance of supporting technologies such as Git for version control, CI/CD for automated deployment, Docker for containerization and Generative AI tools that improve developer productivity.

12. My Understanding

I understood that debugging is an important part of software development. Chrome DevTools and VS Code provide powerful features that help developers quickly identify and solve issues. Learning systematic debugging techniques makes applications more stable and easier to maintain.

13. Conclusion

This module improved my understanding of frontend debugging and taught me how professional developers analyze application behavior instead of simply guessing the cause of errors.